NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.6b Wind-damage loss and wind-speed databases

Group AI Weather & Climate Models for Wind Engineering

Status Shelved

Period 2026-10-14 14:36 — 2026-10-15 02:32

Stage conclusion

Project report

I Wind-damage loss–wind-speed database

Objective

Use large language models to batch-extract NOAA Storm Events narratives (and later post-disaster survey reports) into structured "loss–wind-speed" records, and determine whether this direction is worth pursuing. Liang Hu 10-14: For resilience/performance metrics to be used in design, they must first be validated against real loss data; this is currently difficult, so we are only making an initial start.

Current Status

Shelved (10-14 22:4x), decision. - **Why it does not work**: LLM extraction is usable (source classification 66/70, able to separate wind speeds inferred from damage), but data are lacking—thunderstorm station networks cannot capture damage-core winds (nearby station gusts are a median of 37 mph lower than DAT damage-estimated winds, and H4 reconstruction is ineffective); hurricane DAT contains only damaged points with no trend; CRASAR patches that include a denominator cover small areas with a wind-speed span of only 0.3 m/s and lie entirely within storm-surge zones, leaving 0 usable buildings after excluding water damage. - **What is missing**: Inland building-by-building damage data spanning a wide wind-speed band, excluding water damage, and including undamaged buildings; an independent wind-speed source for thunderstorms. - **Conditions for restarting**: (1) Wide-area post-disaster aerial imagery + building damage classification, systematic inland sampling surveys, or address-level insurance/FEMA data (requires a data agreement, to be decided by Liang Hu); (2) H6 demonstrates that radar near-surface winds can serve as an independent wind speed for thunderstorms; (3) A specific user needs design wind speed vs. actual damage validation and is willing to invest in manual building-by-building verification. Upon restart, reuse scripts and (intermediate data archived on TACC), and run computations on CRC/TACC/Colab rather than locally on the workstation.

Hierarchy Rules (Set by Root Lead on 10-14; 5-level cap after migration)

- Do not open third-level sub-projects under this node; if subdivision is needed, use tasks within this node, or report to the parent node to open a peer second-level node.
- Only two hops remain downward: this node → bot → tool; task briefs must not allow tool bots to delegate further.
- Subtree lists are not grouped by parent node; the UI project tree is authoritative.

Boundaries

- Colab (updated 10-14 18:3x by Parent Project Lead per Liang Hu's exact words "use Colab more at this stage, no CU limit"; original ≤20 CU cap voided): No per-experiment cap; if a single experiment is expected to exceed 100 CU, pause and report to parent node 50648743 first; must run colab stop when finished, no idle running allowed; dispatch only one executor per round.
- Compute locations (tightened by Liang Hu on 10-14 across all tracks): The workstation does not run computations or store compute data. Upon restart, downloads, batch-extraction caches, point-wise reconstruction recalculations, and imagery/raster processing must all be run and stored on CRC/TACC or Colab; the workstation only holds scripts, small results <200 MB per item, figures, and reports, and any temporary storage >1 GB requires reporting to the parent node first. Intermediate data >1 MB previously on the workstation for this node have been moved to TACC by the Root Lead; fetch from

TACC for review and do not pull them back to the workstation. Prioritize TACC for new large data (CRC group storage is at ~95%); CRC defaults to a single node.

- Supercomputer single job ≤50 SU (resource request + registration, pre-check and submission in two separate commands); batch routine tasks ≤100 per batch, no --force, 10% review; data from other tracks are read-only; write only within the designated workspace.

Settled Decisions

- 2026-10-14 Shelved →
- 2026-10-14 After r3: Do not invest further in the DAT × reconstruction sub-route; check public damage data with denominators first before deciding whether to shelve →
- 2026-10-14 After full r2: Put thunderstorms on hold, run hurricane pilot first →
- 2026-10-14 Continue after r1, shifting Step 2 to "co-located, simultaneous pairing feasibility" →
- 2026-10-14 Kickoff design (field schema, six-category wind-speed source taxonomy, sample stratification) →

What Was Tried and Results

Date	What Was Done	One-Line Result
2026-10-14	r1b scoring fix (unit conversion, station semantic tier)	Only changed tropical wind-speed value 17→18 and tropical station semantics 15/20; r1 conclusion unchanged
2026-10-14	r5 supplement: stratify by distance-from-eye RMW bands; change DAT unit to "inferred mph"	All 1,024 buildings across both patches fall in the 1–3 RMW band, with 0 buildings in the <1 and >3 bands, making cross-band comparison impossible; shelved judgment unchanged
2026-10-14	r5 CRASAR small patches (Ian Harlem Heights 912 buildings, Idalia Jena 112 buildings) × centroid point H2 reconstructed gusts	Single-patch gust span in Ian is only 0.3 m/s (< H2 error of 3–4.5 m/s); 0 buildings remain after conservative water-damage exclusion, or 48 buildings across 4 clusters with no major damage after relaxed exclusion; version including water damage also shows no monotonic gradient → Shelved

Date	What Was Done	One-Line Result
2026-10-14	r4 check of public datasets with denominators	Found CRASAR-U-DROIDs (drone imagery building-by-building polygons + no/minor/major/destroyed damage, CC BY 4.0, data card verified by Lead), overlapping with H2's Ian/Ida/Idalia; one Idalia Steinhatchee patch has 608 buildings (473 undamaged); xBD is non-commercial, NSI/OpenFEMA lack building-level damage; Idalia 72–74 m/s (gust; corresponding to sustained wind 55.8–57.1 m/s at 1.10–1.20 RMW) exceeds NHC TCR landfall intensity of 100 kt (PDF p. 5) and has been corrected as **extrapolation distortion** (see correction section in the 10-14 23:0x shelving decision)
2026-10-14	r3c manual check of near-station groups (Helene 5 groups / 78 points, Dallas 3 points)	Helene near-station points are almost entirely tree damage, DAT timestamps are identical in batches (suspected survey entry timestamps), up to 48 points in a single group share one observation, and the nearest model input station for the SBO group is 82.6 km away, exceeding the protocol validation range; none qualify as a co-located, simultaneous ground-truth group
2026-10-14	r3 hurricane pilot: Ida/Idalia/Helene 416 DAT points × point-wise recalculated H2 Holland+residual P2	416/416 reconstructable with complete fields; FR12 89 points show no monotonic trend between DOD and reconstructed gusts (Ida Spearman +0.08, DOD2 accounts for 60/75); main reasons are that DAT contains only damaged points, lacks a denominator, and mixes non-wind mechanisms; Idalia 72–74 m/s pending verification
2026-10-14	r2-C supplement: H4 Dallas 5 days, H2 Idalia/Helene, footprint scale	Dallas yielded candidates only on 2025-03-04: ≤10 km has 3 points / 3 groups, 25 km has 52 points / 9 groups, of which 49 points are 10–25 km from a station, far exceeding downburst scales (microburst <4 km, macroburst ≤10 km: NOAA JetStream; Romanic et al. 2022 WACE 37:100474 journal p. 1); Idalia ≤10 km has 5 points / 3 groups, Helene has 78 points / 5 groups; H2 grid only provides values at station locations, so DAT point-wise reconstruction must be calculated separately

Date	What Was Done	One-Line Result
2026-10-14	r2-C full run: DAT 2019–2024 18,434 points × IEM window maximum gust	Thunderstorms have 667 independent station-hour groups at ≤10 km (1,972 at ≤25 km), tropical has 35 (86); thunderstorm DAT damage-estimated winds are a median of 37 mph higher than nearby station window maximum gusts (10–58, 97% positive; mph unit is assumed) → station networks cannot capture damage-core winds
2026-10-14	r2-B enriched source-positive hard gold standard (70 records)	Six-category source classification 66/70 (94%); measured precision 34/34, recall 34/37; damage-inferred precision 16/17, recall 16/16; unspecified-basis precision 14/17; applies only to enriched positive examples, do not extrapolate occurrence rates; multi-station multi-value records changed to per-value arrays, per-value gold standard not yet built
2026-10-14	r2-C pairing pilot: DAT points × IEM ASOS measured gusts (Ian, Ida, Dallas 2 days, Houston 1 day)	Hurricanes: 131 points yielded 10/47/107 candidates at ≤5/10/25 km; thunderstorms: 112 points yielded 2/71/112; however, independent station-hour groups are very few (~20 groups across 3 thunderstorm days, 28 groups for Ida, 4 groups for Ian); almost none can be assembled at 5 km; DAT built-in wind speeds are damage estimates and do not count as independent
2026-10-14	r1: Storm Events 300 records (thunderstorm 120 / tropical 90 / High Wind 90) Gemini 3.8 Flash extraction + 60 gold standard + 10% review per batch; supplemental batch of 40 TX4 records (2010–2017)	All ok; all three batch review gates ≥90%; strictly independent wind-speed share is 2.5% for thunderstorms, 31% for tropical (48% among those with speeds), 53% for High Wind; EG→MG binary agreement 203/210; supplemental batch structure type 3/4 did not pass the gate; scoring has unit conversion errors pending fix
2026-10-14	r1b: fix scoring (unified mph/kt/m/s conversion, ±0.5 m/s; add mutual-inclusion tier for station semantics), without re-running the model	Tropical wind-speed value 17→18/20; station semantics: thunderstorm 20 / tropical 15 / High Wind 19 (exact-match tier 20/12/18 unchanged); remaining fields unchanged; thunderstorm 20/20 are mostly null-value items and do not represent positive-case capability
2026-10-14	Verify NCEI official format documentation	Four-category definitions of magnitude_type verified; category E does not specify the basis of estimation, requiring narrative inspection

Parameter Source Table

Parameter	Value / Definition	Sources Checked
Idalia landfall intensity	100 kt (1 min sustained wind); maximum measured gust 74 kt (Perry); landfall RMW 10 n mi	Original NHC TCR text
magnitude unit	Wind speed, knots (kt)	NCEI format PDF
magnitude_type	EG estimated gust; ES estimated sustained wind; MS measured sustained wind; MG measured gust	NCEI format PDF
Basis for estimated wind speed (whether estimation from damage is permitted)	Estimating wind speed from damage is permitted	NWSI 10-1605
DAT windspeed unit	mph (inferred from internal data consistency, not officially stated)	DAT layer metadata, DAT data self-check; NWS field dictionary not found
Narrative gives wind-speed range	Record lower and upper bounds; judged correct if falling within the range	—
Texas four-city subset type distribution	EG 450, MG 94, ES 1, MS 1 (546 total)	Data from this track

Open Items and Next Steps

- 10-14 22:3x Parent node supplement: Examine hurricane damage–wind-speed trends separately by distance-from-eye bands (<1 / 1–3 / >3 RMW) (both methods underestimate by ~2 m/s at <1 RMW), already incorporated into Step 5; thunderstorm "radar + station" minimum feasibility check to be conducted by H, and whether to restart will be decided once results are out
- 10-14 22:1x: Step 5 dispatched to ideas: deduplicate CRASAR small patches, exclude water damage/occlusion, reconstruct P2 gusts at building centroids, and bin damage probability by building clusters; prioritize patches away from the coast (Steinhatchee and Cocodrie are both heavy storm-surge disaster zones)
- 10-14 21:5x: Step 4 (0 CU) dispatched to ideas to check whether public hurricane damage datasets containing an "undamaged buildings" denominator (xBD, NSI, OpenFEMA, StEER, NIST) can be matched to H2 events; if available, use in Step 4, otherwise shelve
- Sister node H is running H6 "Minimum feasibility check for thunderstorm radar near-surface winds (Dallas two days)": if feasible, this serves as the restart condition for I's thunderstorm route (using radar near-surface winds as an independent wind-speed source), to be picked up once results are out
- 10-14 21:3x: Step 3 hurricane pilot dispatched to ideas (DAT/StEER damage points × H2 Holland+residual reconstruction × nearby station measurements, 2–3 events, examining damage–wind-speed trends by building type)

- 10-14 21:09: results_r2.md submitted (initial A/B/C version); pairing supplements (window maximum gust, 2019–2024 full-sample independent station-hour group counts, DAT damage-estimated wind vs. measured difference, H4 Dallas days and H2 hurricanes, footprint scale column) are still in the ideas queue → decide whether it is worth continuing once out
- 10-14 20:3x Parent node supplement: All DAT built-in wind speeds are treated as damage-inferred and used solely as damage descriptions; paired wind speeds use only ASOS measurements and H reconstructions; thunderstorm days prioritize the 5 H4 Dallas days (2024-05-28, 06-01, 2025-03-04, 06-09, 07-09, reusing h4/ 1-minute data), and hurricanes prioritize H2 events with many land stations; add a "distance vs. footprint scale/RMW" note to the pairing table → incorporated
- 10-14 20:2x: r2 supplemental dispatch—take station window maximum gusts, estimate independent station-hour group counts (true N) across the full 2019–2024 DAT, and compute the distribution of differences between DAT damage-estimated winds and nearby station measurements; 70-record hard gold standard batch is running
- 10-14 17:31 Progress: All 300 main sample records extracted, Batch 3 review passed the gate (structure/severity/quote/source each 9/10); Batch 4 (40 records) is running. Preliminary signals from draft analysis.json (not finalized): among 120 thunderstorm records, 117 are EG, only 3 can be confirmed as measured from the narrative, and 116 are "estimated without stating basis"; among 90 extratropical records, 48 are measured; among 90 tropical records, 54 have wind speeds in the narrative and 26 are measured. Gold standard: thunderstorm fields are near full marks but skewed easy because narratives largely lack wind speeds; tropical structure type (65%), station (60%), and source (75%) are on the low side
- 10-14 16:51 Progress: Batch 2 100/100 completed, 10% review achieved ≥9/10 on each field (structure type and gust/sustained each 9/10) and 8/10 across all fields jointly, passing the gate; Batch 3 started around 16:5x; followed by Batch 4 and results_r1.md
- 10-14 16:19 Progress: Batch 1 100/100 completed, 10% review achieved 9/10 full-field agreement (the sole discrepancy was damage severity moderate vs. severe, leaving that field at 90%, which did not trigger the stop line; all 10 reviewed records happened to be thunderstorms without narrative wind speeds, offering weak evidence for the source field, so the gold standard takes precedence); Batch 2 started at 16:18
- 10-14 15:19 Progress: Batch 1 has produced 31/100, Gemini 5-hour pool has 26% remaining, paused per platform yield rules and expected to auto-resume at 15:53 (did not use --force); Batch 4 (40 TX4 records from 2010–2017) is ready; ideas scheduled to check back around 15:47. Batch sequence: Batch 1 → 10% review gate → Batch 2 → Batch 3 → Batch 4 → results_r1.md
- 10-14 14:50 Verdict: Review pause applies only to the Lab's own reviews; S1 overlap conflicts with the 2018–2023 window (S1 is entirely in 2010–2017) → add supplemental Batch 4 (40 TX4 thunderstorm records, listed separately); thunderstorm wind speeds are mostly in the MAGNITUDE column, so report narrative vs. structured-column coverage separately
- Main sample ready: 300 records (thunderstorm 120 / tropical 90 / extratropical 90, seed 20261014), 60-record gold standard manually labeled (Lead spot-checked 3 records as reasonable); DAT has public ArcGIS layers and exports, while StEER is on DesignSafe/Fulcrum with no unified API and licenses to be verified per project
- ideas Step 1 results → Lead spot-checks gold standard and inspects failure cases → judge whether it is worth continuing

- 10-14 H3 extratropical cyclone official version (lines 108–110; the earlier "2 events, 151 stations, −5.5%" was a voided preliminary run that mistakenly treated METAR TSNO as thunderstorm and is no longer cited): strictly compliant 4 events and 359 stations, peak RMSE Kriging 2.30 → ERA5+residual 2.22 (−3.2%), interpolation is sufficient → for I's extratropical high winds: Ordinary Kriging serves as the baseline, with peak underestimation handled separately.
- 10-14 H4 thunderstorm (Dallas 4 days, 44 1-minute stations, very small sample): 0 gain from fusion; peak sustained wind underestimated by ~3.2 m/s, gust by ~5 m/s, and standalone ERA5 underestimated by ~8.8 m/s → station reconstruction is essentially powerless for thunderstorm high winds. Combined with I Step 1 (only 3 of 120 thunderstorm records have confirmable measurements), thunderstorm events currently have no usable independent wind-speed source.
- H's four-category results compiled into "Station Wind-Speed Reconstruction Protocol v1", to be formally handed over to I after review
- 10-14 H protocol upgraded to **v1.2** (replacing v1.1; Section 4 basis for peak correction uses leave-one-event-out, Section 4 not separately submitted for review): additive correction by event type—benign sustained +0.82 (P2 definition, do not mix with P1 −1.6 to −2.0), extratropical sustained +1.46, extratropical gust +1.38, thunderstorm sustained +3.25, thunderstorm gust +5.08; no correction for hurricane Holland+residual (if taking maximum of Kriging time series, +1.21). Caveats: must be applied by type and cannot be generalized across types; single-event residual bias for thunderstorms can still reach ±2–3 m/s, and the correction magnitude comes from only 4 Dallas events as a rough estimate; label corrected values as "reconstructed, unmeasured, corrected (v1.2)" while keeping raw reconstructed values in a separate column; verified only at station locations, whereas stationless loss points and out-of-window peaks remain unvalidated. Impact on I: r3 hurricanes used Holland+residual, which requires no correction under v1.2, so r3 conclusions are unchanged; if corrected values affect any judgment in the future, ask the parent node to submit Section 4 for review first
- 10-14 H "Station Wind-Speed Reconstruction Protocol v1.1" formally delivered (since replaced by v1.2) (3/3 support with reservations)—**this is authoritative, superseding the interim H numbers below**: reconstructed value records must include method, background field, peak definition P1/P2 and window, sustained/gust and averaging period, distance to station, event type, hurricane RMW band, and subset sample RMSE/bias; peaks are generally underestimated (benign 1.6–2.0, extratropical 1.3–1.5, hurricane interpolated strong-wind stations 2.6 / near-core 2.0, thunderstorm sustained 3.3 / gust 5.1 m/s), so directly building a loss–wind-speed relationship would bias the wind-speed axis low; for hurricane maximum winds, use Holland+residual (strong-wind stations 3.45/−0.66 vs. Kriging 3.87/−2.59), though both underestimate by ~2 m/s at <1 RMW; locations far from stations and widespread station outages remain unvalidated; H5 peak correction is in progress. Written into the supplemental section of the Step 2 task brief: count reconstructed pairings separately from measured pairings, do not include them in the strictly independent tier, and do not use reconstruction for thunderstorms
- 10-14 H hurricane-stage finalized recommendations to I for hurricane wind speeds: (1) If only event maximum sustained wind is needed → Node B Holland wind field + station residual Kriging (peak bias −1.18 → ~0 m/s; across 273 strong-wind stations ≥12 m/s, RMSE 3.89 → 3.50, −10.0%, right on the lower bound of "conditional"; Lead checked results_h2b.md line 39: only 154/273 stations improved); (2) If hourly time series or non-peak wind speeds are needed → Ordinary Kriging (Holland+residual is ~20% worse hourly); (3) Near land stations, interpolation and fusion are comparable overall (−3.2%), while

offshore/coastal platforms cannot be evaluated; (4) For gust peaks, Holland+residual reduces RMSE from 3.67 to 3.47, where the gust conversion factor is an assumption (see Node B parameter source table). Fields remain as before: reconstruction_method, nearest_station_km, averaging (sustained/gust), wind_speed_source=reconstructed (unmeasured). Extratropical cyclones (H3) running, thunderstorms unvalidated

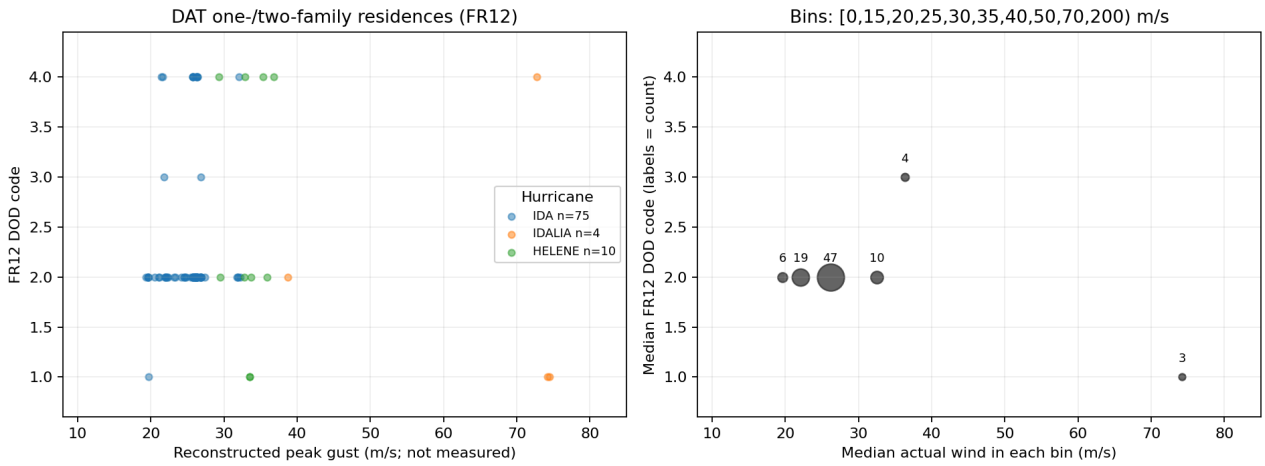
- 10-14 H method recommendations (narrowed after review verdict): Limited strictly to sustained winds under benign wind conditions (ASOS ~2 min average)—use Ordinary Kriging when <50 km from the nearest station, and ERA5+residual Kriging when farther; gusts, peaks, and thunderstorms/tropical cyclones/extratropical cyclones are all unvalidated, and peak hours are generally underestimated by ~2 m/s. Must not be written as "interpolation meets I's needs." Numbers from cleaned finalized version (lines 11–12): in the 82 km band, ERA5+residual Kriging lowers RMSE by 12.47% over the best pure interpolation with 124/134 stations improved, judged "conditional"; in the 54 km band, it lowers RMSE by 9.36%, judged "interpolation sufficient"; on the dense network, it is only 2.6% lower; r1's ">50 km −17%/−24%" came mainly from offshore platform station BQX, dropping to only −8.9% after removal, and is no longer cited. ERA5 does not directly assimilate land 10 m winds (Hersbach 2020, per H verdict) and is largely independent of land target stations. When used by I, label wind_speed_source=reconstructed (unmeasured), and additionally record reconstruction_method, nearest_station_km, and averaging (sustained/gust)
- 10-14 H reminder (line 31): Reconstructed wind speeds systematically underestimate the strongest hour of the day at a station by ~1.5 m/s (best GBT+residual OK bias −1.52 m/s across 1,349 station-days), with larger errors at >50 km from the nearest station → when using reconstructed wind speeds as an independent source in the future, add fields reconstruction_method and nearest_station_km, and treat peaks as biased low (whether to apply bias correction will be decided in Step 2); H's Texas station table and Storm Events 2020–2021 detail files are available for read-only reuse (superseded by the post-review wording in the item above; numbers in the item above govern)
- With H: H's reconstructed wind fields will serve as this node's "independent wind-speed source" going forward; prioritize Texas four-city events in the sample for easier alignment

Literature Basis

- No cards yet; after Step 1 returns, delegate to litwiki as needed to build literature cards for NWSI 10-1605 and damage–wind-speed circularity. No literature basis; based on data from this track.

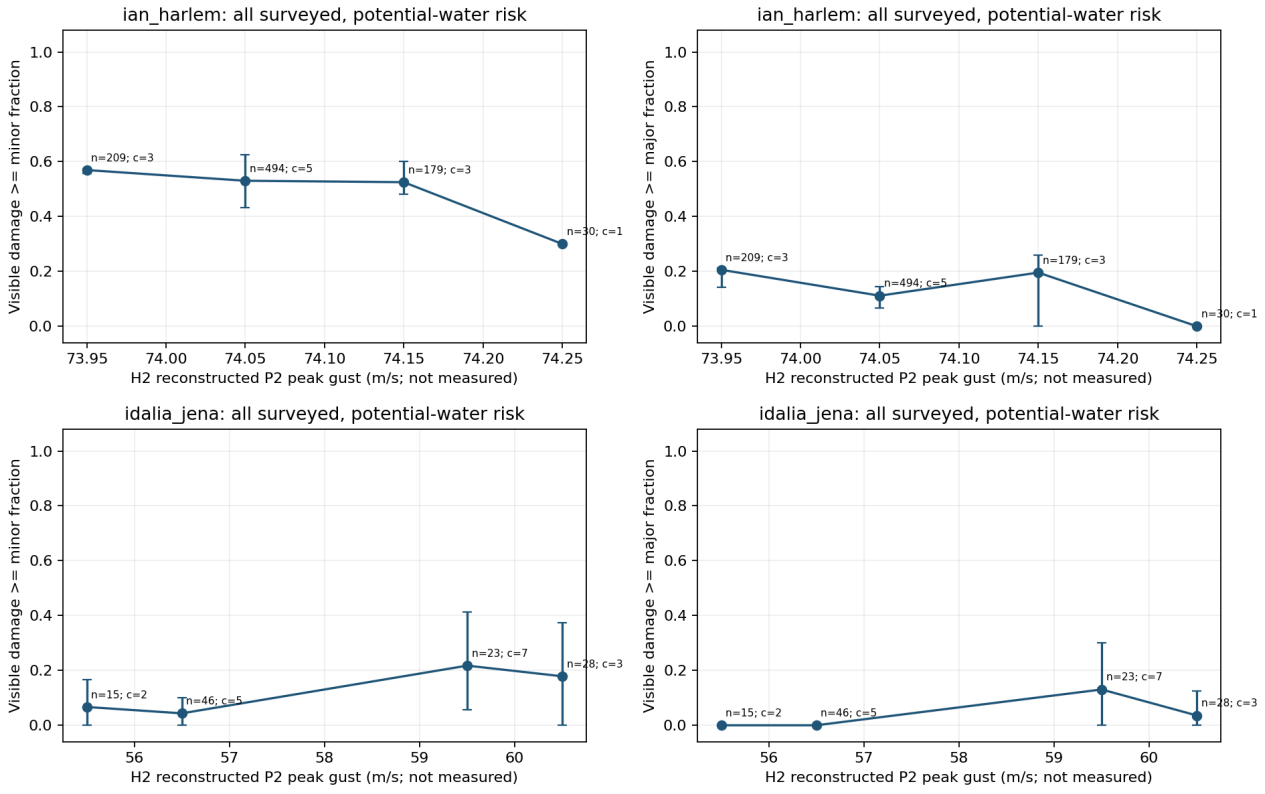
Notes from Liang Hu

Key figures



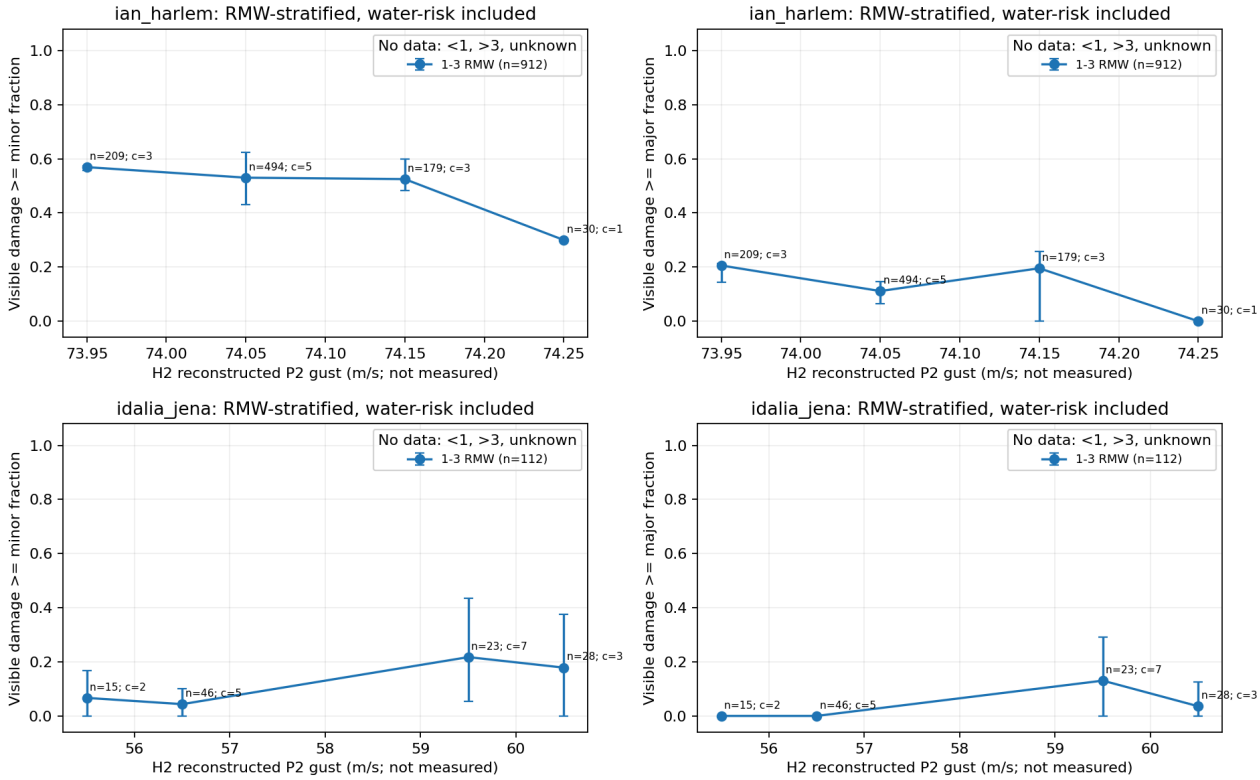
NWS DAT damage points + H2 NHC/ATCF/IEM wind reconstruction; peaks may be underestimated. Damaged-only selection, not a probability curve.

fr12 trend r3b



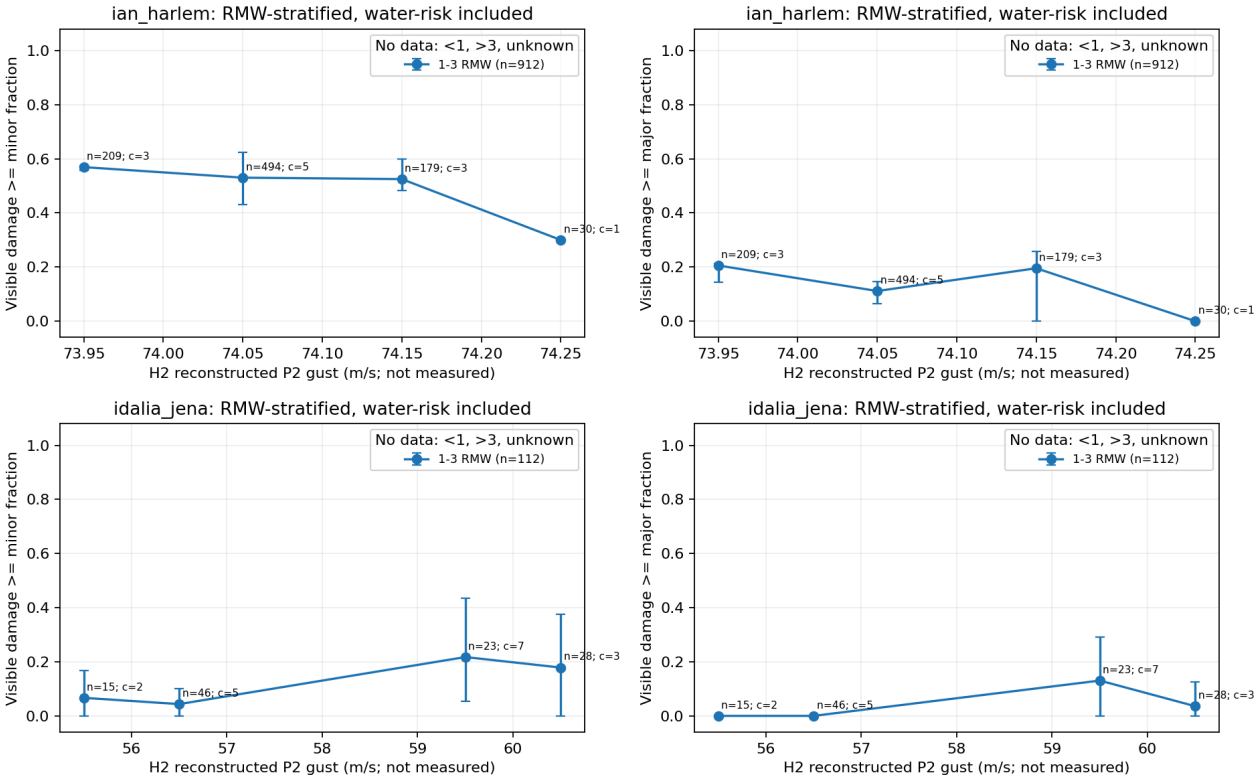
CRASAR-U-DROIDs labels CC BY 4.0; H2 Holland+residual / NHC track; USGS 3DEP and NHC potential surge screen.
All bins include suspected water exposure; reconstructed gust is not measured, and event/point peak may be biased.

r5 damage probability exploratory



CRASAR-U-DROIDs (CC BY 4.0); H2 Holland+residual / NHC best track. All usable buildings lie in 1–3 RMW; no <1 or >3 comparison.
Suspected water exposure included; no water-screened sample. Reconstructed wind may be biased; <1 RMW station validation underestimates by ~2 m/s (not observed here).

r5 damage probability rmw supplement

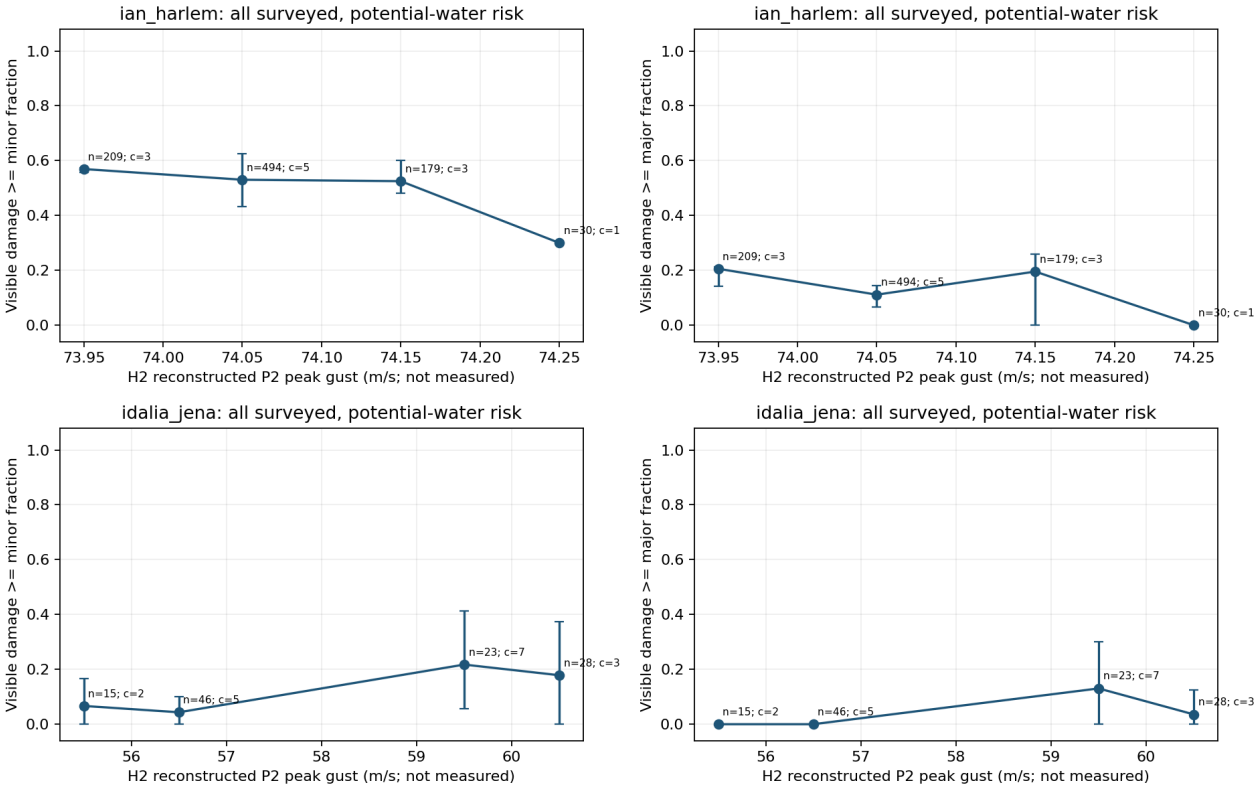


CRASAR-U-DROIDs (CC BY 4.0); H2 Holland+residual / NHC best track. All usable buildings lie in 1–3 RMW; no <1 or >3 comparison.
Suspected water exposure included; no water-screened sample. Reconstructed wind may be biased; <1 RMW station validation underestimates by ~2 m/s (not observed here).

CORRECTION: Ian Harlem Heights (1.02-1.05 RMW) and Idalia Jena gusts are near-core /
not observationally validated at the buildings. Near-core bin: UNVALIDATED; exclude
from primary wind-damage gradient inference. Existing plotted counts remain archival.
After conservative water-risk exclusion: 0 eligible buildings; no primary gradient.

Sources: CRASAR-U-DROIDs (Manzini et al., 2024, arXiv:2407.17673; CC BY 4.0);
H2 reconstruction (Lab H); NHC best track + potential-inundation map; USGS 3DEP.

r5 damage probability rmw supplement annotated



CRASAR-U-DROIDs labels CC BY 4.0; H2 Holland+residual / NHC track; USGS 3DEP and NHC potential surge screen.
All bins include suspected water exposure; reconstructed gust is not measured, and event/point peak may be biased.

CORRECTION: Ian Harlem Heights (1.02-1.05 RMW) and Idalia Jena gusts are near-core /
not observationally validated at the buildings. Near-core bin: UNVALIDATED; exclude
from primary wind-damage gradient inference. Existing plotted counts remain archival.
After conservative water-risk exclusion: 0 eligible buildings; no primary gradient.

Sources: CRASAR-U-DROIDs (Manzini et al., 2024, arXiv:2407.17673; CC BY 4.0);
H2 reconstruction (Lab H); NHC best track + potential-inundation map; USGS 3DEP.

r5 damage probability exploratory annotated